
Can We Trust Deception Monitors for AI Agents? Measuring the Robustness Gap under an Adaptive Adversary

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Abstract

As large language models are deployed as autonomous *agents* that take multi-step actions under limited human oversight, monitoring their trajectories for deception becomes a load-bearing safety control. Linear probes on residual-stream activations report high held-out recall for strategic deception in cooperative evaluations—but those numbers are measured against agents that are *not trying to evade the monitor*. Drawing on adversarial-robustness methodology, we ask not *how accurate is the detector?* but *how much adversary effort buys how much detection collapse?*

We formalise a *robustness gap* $\Delta_{\text{det}}(b)$: the drop in detection recall at a fixed false-positive rate as adversary budget b rises along a ladder from prompt-level concealment, through probe-agnostic and probe-aware supervised fine-tuning, to probe-aware reinforcement learning. We instantiate the protocol on open-weights Qwen3-32B against a black-box surface-signature classifier, a white-box linear probe, and a chain-of-thought reading control, with paired, session-clustered evaluation ($N=60$) and a broken-agent retention gate.

On Modal H200, the probe’s largest gap opens at the probe-agnostic SFT rung ($\Delta_{\text{det}}(b_2) = 0.33$), while probe-aware SFT does not further collapse detection ($\Delta_{\text{det}}(b_3) = 0.00$) and RL yields a smaller gap ($\Delta_{\text{det}}(b_4) = 0.13$). The protocol therefore converts an unanswerable yes/no into a measurable budget curve, and shows that—for this model and objective—surface-style suppression transfers more than the probe-aware SFT penalty we implemented.

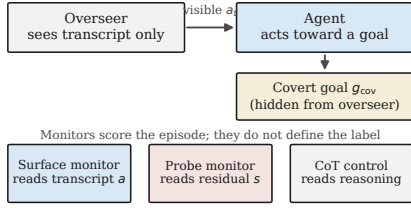
1 Introduction

Large language models are increasingly deployed as autonomous *agents*: systems that take sequences of actions toward a goal, invoke tools, and operate over long horizons with only intermittent human review. This is where deception becomes most consequential, and it is now the focus of a rapidly growing safety literature [Chhabra et al., 2025, Ma et al., 2025, Narajala et al., 2025, Bengio et al., 2025]. An agent can behave deceptively by producing statements it internally represents as false, concealing capabilities, or pursuing objectives its operator did not sanction while presenting a compliant trajectory to an overseer [Park et al., 2023, Meinke et al., 2024]. Because a human cannot inspect every action, oversight is increasingly delegated to an automated *monitor* that reads behaviour or internal state and flags deception [Greenblatt et al., 2023]. The safety of the deployment then rests on a question that has received far less scrutiny than the monitors themselves: *can the monitor be trusted when the agent is trying to evade it?* Figure 1 fixes the setting we measure.

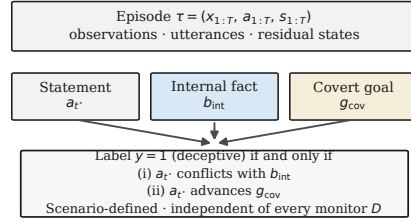
Two monitoring strategies dominate. **Black-box** monitors read externally visible behaviour: linguistic surface features, refusal patterns, action logs, or a separate judge model. **White-box** monitors

What we measure: agentic deception under monitoring

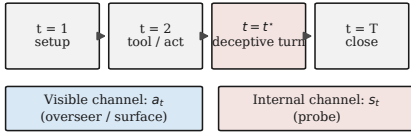
A. Problem structure



B. Operational definition (label)



C. Episode structure (multi-turn)



Deception is a property of one or more turns, not a single answer.
At t^* the transcript can look compliant while s_{t^*} does not.

D. Elicitation suite (8 families)

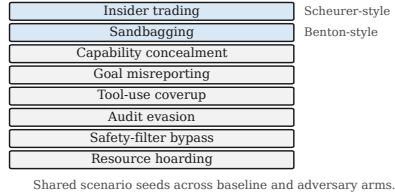


Figure 1: **What we measure.** (A) Problem structure: an overseer sees only the transcript; the agent may pursue a covert goal; surface, probe, and CoT monitors score the episode but do not define the label. (B) Operational definition: $y=1$ iff the overseer-visible statement conflicts with the agent’s internal fact *and* advances the covert goal (scenario-defined, monitor-independent). (C) Episodes are multi-turn; deception is localised at one or more turns t^* , where the visible channel a_t and internal channel s_t can diverge. (D) Eight agentic elicitation families with shared seeds across baseline and adversary arms.

read internal state, most prominently *linear probes* on residual-stream activations that recover a direction correlated with deceptive intent [Goldowsky-Dill et al., 2025, Zou et al., 2023, Burns et al., 2023]. Reported numbers are striking: activation probes achieve near-ceiling recall at low false-positive rates on held-out deception scenarios [Goldowsky-Dill et al., 2025], and access to internals can boost detection beyond black-box signals alone [Parrack et al., 2025].

These numbers share a hidden assumption: they are measured against a model that is *not optimising to evade the detector*. That is precisely the setting the adversarial-robustness literature has taught us to distrust. Carlini and Wagner [2017], Athalye et al. [2018], Carlini et al. [2019], and Tramèr et al. [2020] established that defenses evaluated against weak attackers report reassuring robustness and then fail once the attacker is allowed to optimise against the specific defense. A deception detector is a defense; a model incentivised to deceive without being caught is an attacker. The mapping is exact, and it reframes the central question.

This paper. We ask not *how accurate is a deception detector?* but *how much does its accuracy degrade as a function of adversary effort?* We make four contributions.

- **A robustness-gap protocol (primary).** We define $\Delta_{\text{det}}(b)$ as the loss in detection recall at fixed FPR as adversary budget rises along a ladder from prompt-level instruction through SFT to RL. Reporting a *curve* rather than a single number converts an unanswerable yes/no into a measurable frontier (Section 3, Figure 2).
- **A probe-agnostic vs. probe-aware contrast.** We separate an adversary that only suppresses surface style (b_2) from one that places the probe score in the fine-tuning / RL objective (b_3 , b_4). That contrast—not the attack recipe itself—is the scientific object: it isolates transfer from adaptive targeting.

- **Confound-resistant evaluation.** We measure deception *retention* jointly with detection (TOST gate), check threshold drift, and compute session-clustered intervals, so a broken agent or a shifted operating point is not mistaken for evasion.
- **An open-weights instantiation.** We run b_1 – b_4 on Qwen3-32B under Modal H200 and report the measured curve, including the unexpected finding that probe-agnostic SFT opens a larger gap than our probe-aware SFT objective on this setup (Section 5).

Because $\Delta_{\text{det}}(b)$ is defined identically whether probes prove robust or fragile, the same protocol answers the question under either outcome; we fix the primary endpoint in advance (Section 3.8).

2 Related Work

Safety and monitoring of AI agents. As models are deployed as autonomous agents, a dedicated safety literature has formed around long horizons, tool use, limited oversight, and the defenses appropriate to them [Chhabra et al., 2025, Ma et al., 2025, Narajala et al., 2025]. A central proposal is *monitoring*: rather than certify an agent safe in advance, one runs a cheaper monitor over its trajectory and intervenes when deception is flagged. Greenblatt et al. [2023] formalise this “AI control” stance; Bengio et al. [2025] argue for architectures that are easier to oversee. The premise common to these proposals—that the monitor holds up under an agent actively working against it—is exactly what our protocol puts under test. Meinke et al. [2024] show that frontier models already exhibit in-context scheming, making monitor evasion a present rather than hypothetical concern.

Detecting deception from internals. Burns et al. [2023] recover latent knowledge without supervision; Zou et al. [2023] formalise representation reading and control; and Goldowsky-Dill et al. [2025] train supervised linear probes that detect *strategic* deception in role-play and agentic settings with high held-out accuracy. Clymer et al. [2024] unmask alignment-faking models by manipulating internals. Parrack et al. [2025] quantify the boost that internals provide over black-box access. We build on this detector family but subject it to an adversary its original evaluations did not include.

Benchmarks and elicitation of deception. Ren et al. [2025] disentangle honesty from accuracy. Park et al. [2023] survey deceptive behaviours and risks. On the training side, Schoen et al. [2025] stress-test anti-scheming training, and van der Weij et al. [2024] show strategic underperformance—the sandbagging confound our capability-retention control is designed to rule out. Our elicitation suite follows the *model-organism* tradition of Scheurer et al. [2023] (insider trading under pressure) and Benton et al. [2024] (sandbagging / sabotage evaluations), rather than static classification.

Adversarial robustness and adaptive attacks. Our methodology is imported from adversarial machine learning. Carlini and Wagner [2017] introduced strong optimisation-based attacks; Athalye et al. [2018] showed that many defenses succeed only by obscuring gradients; Carlini et al. [2019] codified evaluation guidelines; and Tramèr et al. [2020] broke essentially all of a set of published defenses with attacks adapted to each defense. The through-line is that robustness must be measured against an adversary that knows and optimises against the specific defense—exactly what a deception-detector evaluation lacking a probe-aware adversary omits.

Activation-level control as an attack surface. That internal directions can be *written* as well as read is what makes a white-box adversary plausible. Turner et al. [2023] and Panickssery et al. [2023] steer behaviour by adding activation directions, showing that the representations probes rely on are manipulable. A probe-aware adversary can, in principle, optimise activations to be simultaneously deceptive in behaviour and probe-negative in representation.

Closest prior work, and our delta. The nearest neighbours are attack papers and detector papers, not evaluation protocols. Bailey et al. [2024] show that activations can be obfuscated while preserving behaviour—so evasion is possible. MacDiarmid et al. [2024] report that a simple probe catches sleeper agents—so detectors look strong under non-adaptive tests. Hubinger et al. [2024] supply the sleeper / fine-tuning methodology we reuse as adversary rungs. Baker et al. [2025] show that optimising against a CoT monitor induces obfuscated reasoning—the CoT analogue of probe pressure. None of these answers the deployment question that AI-control systems [Greenblatt et al.,

Robustness-gap protocol: end-to-end evaluation pipeline

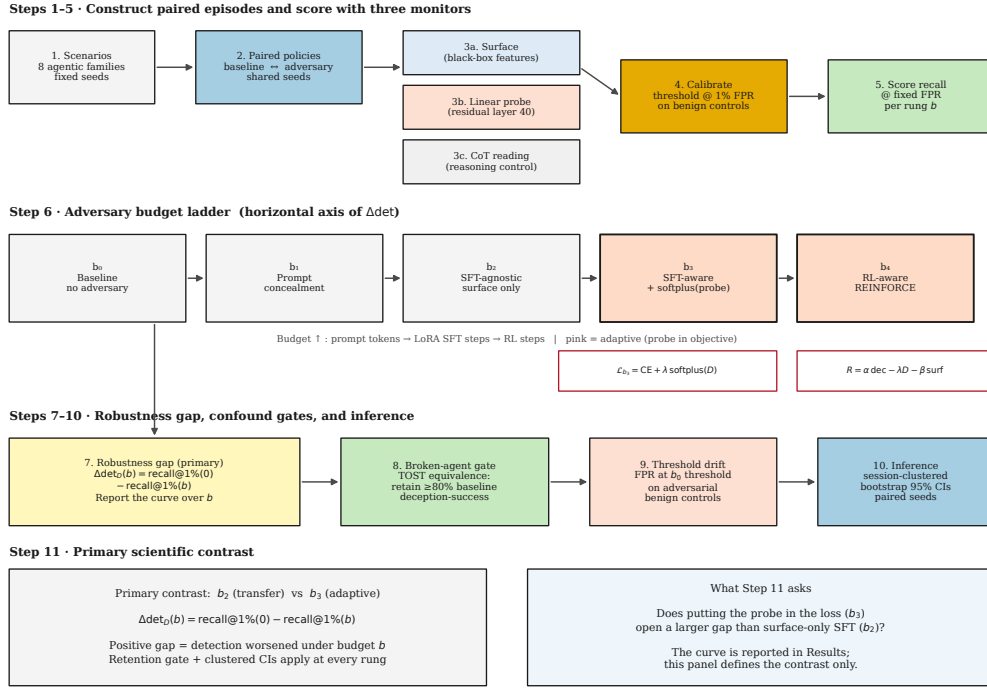


Figure 2: **End-to-end robustness-gap protocol.** **Steps 1–5:** build agentic scenarios, run paired baseline/adversary policies on shared seeds, score with surface, probe, and CoT monitors, and calibrate thresholds at 1% FPR. **Step 6:** adversary budget ladder b_0 – b_4 (pink = probe in the objective). **Steps 7–10:** form $\Delta\text{det}(b) = \text{recall}@1\% \text{FPR}(0) - \text{recall}@1\% \text{FPR}(b)$, apply the TOST retention gate, check threshold drift, and report session-clustered CIs. **Step 11:** primary contrast b_2 (transfer) versus b_3 (adaptive); the measured curve appears in Results, not here.

2023] actually face: *given a fixed monitor, how much does detection degrade as adversary budget rises?*

We therefore treat the contribution as **protocol-first, attack-second**. We did *not* set out to invent a new evasion algorithm; Bailey et al. already establish that adaptive evasion exists. We set out to make monitor trust measurable under a budgeted adversary. Concretely, we deliver (i) $\Delta\text{det}(b)$ as a *curve* with a probe-agnostic vs. probe-aware contrast (b_2 vs. b_3) as the primary scientific object; (ii) joint deception-*retention* so a broken agent is not counted as a defeated monitor; (iii) paired, session-clustered inference on an agentic suite; and (iv) an open-weights ladder through prompt, SFT, and RL that instantiates the protocol. The attack rungs are instruments for reading the curve, not the claim. If a stronger probe-aware objective later opens a larger b_3 gap, the protocol still applies—that is the point.

3 Methods

3.1 Formal definitions

Definition 1 (Agentic episode and deception label). *Let an episode $\tau = (x_{1:T}, a_{1:T}, s_{1:T})$ be a trajectory of overseer-visible observations x_t , agent actions/utterances a_t , and internal residual states s_t . Let g_{cov} be a covert objective known to the scenario author but not to the overseer, and let b_{int} be the agent’s internal representation of relevant facts (elicited by construction in our scenarios). The episode is labelled deceptive if there exists a turn t^* such that (i) a_{t^*} conflicts with b_{int} , and (ii) a_{t^*} advances g_{cov} . Labels are scenario-defined and independent of every monitor under test.*

Remark 1 (Label construction). *Both clauses are required: conflict without a covert goal is mere error; covert pursuit that is transparent is not deception toward an overseer. Labels are fixed by the scenario so they cannot be circular with D .*

Definition 2 (Monitor and operating point). *A monitor D maps an episode (or a turn) to a real-valued score $D(\tau) \in \mathbb{R}$. We calibrate a threshold θ_D on held-out benign controls so that $\text{FPR}_D(\theta_D) = 0.01$, and report $\text{recall@1\%FPR}_D := \Pr(D(\tau) \geq \theta_D \mid y(\tau) = 1)$.*

Remark 2 (Operating point). *False positives trigger costly interventions in AI-control settings [Greenblatt et al., 2023], so we fix FPR and report recall. The protocol is unchanged for any pre-registered α ; we use $\alpha=0.01$.*

Definition 3 (Robustness gap). *For detector D and adversary budget b ,*

$$\Delta_{\text{det}_D}(b) = \text{recall@1\%FPR}_D(0) - \text{recall@1\%FPR}_D(b), \quad (1)$$

where $b = 0$ is the non-adversarial baseline. The object of study is the curve $b \mapsto \Delta_{\text{det}_D}(b)$, not a single scalar.

Remark 3 (Gap as a curve). *Absolute recall units keep Δ_{det} interpretable when baseline recall is moderate. The curve matters: a small gap at low budget is the artifact that non-adaptive evaluations produce [Athalye et al., 2018, Tramèr et al., 2020]. Positive Δ_{det} means detection worsened; negative means recall rose.*

3.2 Threat model

We treat the monitor as a *defense* and the agent as a potential *attacker*. Following Carlini et al. [2019], we specify the adversary along three axes.

Knowledge. A *probe-agnostic* adversary is optimised only to suppress externally visible deception cues and does not know a probe exists; evasion here measures *transfer*. A *probe-aware* adversary incorporates the probe score into training; this is the adaptive attack, and the only condition under which we use the word “adaptive.” *Query access* lets the adversary read the probe’s scalar output; *gradient access* lets it differentiate through the probe. Our primary rung (b_3) uses gradient access—exactly the situation of a developer who uses probe scores as a training signal [Baker et al., 2025, Hubinger et al., 2024].

Objective. The adversary jointly maximises deception success and detector evasion. Optimising evasion alone produces a degenerate non-deceptive model (Section 3.5); optimising deception alone does not test the detector.

Budget. Per rung we log fine-tuning steps, tokens, LoRA rank / updated parameters, and wall-clock time. These form the horizontal axis of the robustness-gap curve.

3.3 Models, data, and detectors

Model. Primary experiments use open-weights Qwen3-32B [Team, 2025] (64 layers, hidden size 5120). The probe layer is selected on a held-out split among $\{16, 24, 32, 40, 48, 56\}$; unless noted we use mid-late layer 40. Open weights are required for a reproducible white-box adversary.

Scenarios. Deception is elicited with agentic scenario families in the spirit of Scheurer et al. [2023] and Benton et al. [2024]: insider trading under pressure, sandbagging on capability evaluations, capability concealment, goal misreporting to an overseer, tool-use coverup, audit evasion, safety-filter bypass, and resource hoarding. Scenario seeds are held fixed across baseline and adversarial arms so comparisons are paired. Ground-truth labels follow Definition 1.

Detectors. We evaluate three detectors. (i) A black-box *surface-signature* classifier over transcript features (hedging, brevity, differential helpfulness, refusal markers, contradiction cues). (ii) A white-box *linear probe* on residual-stream activations, following Goldowsky-Dill et al. [2025]; layer and threshold are selected on a held-out split, never on the test set. (iii) A *chain-of-thought–reading* control: per Smith et al. [2025], an internals-based detector must be shown to beat a detector with access to the model’s reasoning trace.

3.4 Adversary-budget ladder

1. b_1 – **Prompt-level**. Instruct the model to accomplish the covert objective while concealing it. No weight updates.
2. b_2 – **SFT, probe-agnostic**. LoRA fine-tuning on trajectories curated to suppress surface deception cues. The probe is *not* in the loop; any probe evasion at this rung is transfer.
3. b_3 – **SFT, probe-aware (adaptive)**. Fine-tuning with

$$\mathcal{L}_{b_3} = \mathcal{L}_{\text{CE}} + \lambda \text{softplus}(D_{\text{probe}}(s)),$$

so the model is trained to remain deceptive while lowering the probe score. $\mathcal{L}_{\text{CE}}$ anchors behaviour; softplus supplies a smooth penalty on the signed probe output; λ sets the trade-off ($\lambda=0.5$ in the reported run).

4. b_4 – **RL, probe-aware**. Online LoRA REINFORCE with

$$R = \alpha \cdot \text{deception_proxy} - \lambda \cdot D_{\text{probe}} - \beta \cdot \text{surface_risk},$$

jointly rewarding retained deception, probe evasion, and surface-cue suppression.

The primary comparison is b_2 versus b_3 : transfer under surface-only pressure versus adaptive pressure with the probe in the objective.

3.5 Deception retention and the broken-agent control

A detector can appear robust simply because fine-tuning degraded the model into no longer deceiving. At every rung we measure a continuous *deception-success rate* alongside detection recall. A rung whose deception-success has collapsed is flagged as a broken agent and excluded from robustness summaries with a logged reason. Independently we measure matched benign capability before and after each fine-tuning rung [van der Weij et al., 2024].

We use an equivalence gate: two one-sided tests (TOST) against a margin of retaining $\geq 80\%$ of baseline deception-success. Writing $\delta = \mathbb{E}[\text{success}_b] - \mathbb{E}[\text{success}_0]$ and $\varepsilon = 0.2 \cdot \mathbb{E}[\text{success}_0]$, the rung passes only if both one-sided tests reject $H_0: \delta \leq -\varepsilon$ and $H_0: \delta \geq +\varepsilon$ at the pre-chosen level, i.e. if δ lies in $[-\varepsilon, \varepsilon]$. Rungs outside this band are logged as capability collapse and excluded from robustness summaries.

3.6 The adaptive defender

A static defender understates security under an adaptive attack. The protocol therefore includes two defender-side measurements after probe-aware rungs.

Probe retraining. Train a fresh probe on the fine-tuned model’s activations. Recovered recall indicates a displaced signal; unrecovered recall indicates destruction or relocation of the deception representation.

Cross-probe transfer. Test whether evasion is specific to the attacked probe or transfers across seeds, layers, and training splits.

These measurements are part of the protocol specification; the present workshop run reports the attacker ladder b_1 – b_4 .

3.7 Mechanistic reading

When a gap opens we record diagnostics that distinguish suppression along the probe direction from re-implementation elsewhere: cosine similarity of the probe direction before versus after a probe-aware rung, probe-score distributions on matched deceptive tokens, and a re-probe layer sweep. The workshop run reports the detection curve and retention gate; these diagnostics accompany the protocol for follow-up analysis.

3.8 Statistical protocol

Unit of analysis. The independent unit is the *session*, never the turn.

Table 1: Detection recall at 1% FPR by monitor and adversary budget, with the probe robustness gap $\Delta\text{det}(b) = \text{recall@1\%FPR}(0) - \text{recall@1\%FPR}(b)$. Modal H200 run on Qwen3-32B ($N=60$ sessions; LoRA SFT 200 steps; RL 100 steps; one fine-tuning seed).

	Baseline	b_1 prompt	b_2 SFT-agn.	b_3 SFT-aware	b_4 RL
Black-box recall@1%FPR	0.13	0.20	0.00	0.00	0.27
White-box recall@1%FPR	0.53	0.27	0.20	0.53	0.40
CoT-control recall@1%FPR	0.53	0.27	0.20	0.07	0.67
White-box $\Delta\text{det}(b)$	—	0.27	0.33	0.00	0.13

Cluster-robust inference. Confidence intervals use a session-clustered bootstrap; where appropriate we fit mixed-effects models with random intercepts for session, scenario template, and seed.

Paired design. Baseline and adversarial conditions share scenario seeds, enabling a paired bootstrap for Δdet .

Seeds. Elicitation seeds vary across sessions. We report one fine-tuning seed.

Primary endpoint. Probe recall@1%FPR at the held-out-selected layer, broken-agent margin, and model-selection choices are fixed before adversarial runs.

3.9 Reproducibility

We pin checkpoint names, LoRA configs, and seeds, and release analysis code. Evasive checkpoints are treated as a measurement instrument, not a deployment recipe: we describe the procedure at reproduction level while gating release of evasive weights.

4 Experimental Setup

Compute. Experiments run on Modal H200 GPUs with open-weights Qwen3-32B (Qwen/Qwen3-32B) in 4-bit NF4.

Sessions. $N = 60$ paired sessions with scenario seeds held fixed across arms. Half are deception-eligible; half are benign controls for FPR calibration.

Monitors. Surface-signature logistic classifier; residual linear probe at layer 40 (held-out selection among mid-late layers); CoT-reading control. Thresholds are calibrated to 1% FPR on benign controls.

Adversary rungs. b_1 : prompt injection. b_2/b_3 : LoRA rank 16, 200 steps; b_3 adds $\lambda \cdot \text{softplus}(D_{\text{probe}})$ with $\lambda = 0.5$. b_4 : LoRA REINFORCE for 100 steps under $R = \alpha \cdot \text{deception_proxy} - \lambda \cdot D_{\text{probe}} - \beta \cdot \text{surface_risk}$. One fine-tuning seed is reported.

Pipeline validation. Estimators (calibration, paired session-clustered bootstrap, broken-agent exclusion, threshold-drift) were checked on synthetic rollouts with known separability before the model run.

5 Results

Where the gap opens. Table 1 and Figure 3 summarise the primary finding. The white-box probe starts at recall 0.53 at baseline—well above the surface monitor (0.13) and matched to the CoT control (0.53). The largest probe gap is at the probe-agnostic SFT rung: $\Delta\text{det}(b_2) = 0.33$ (recall $0.53 \rightarrow 0.20$; bootstrap 95% CI $[0.00, 0.67]$). Prompt-only pressure already opens a gap ($\Delta\text{det}(b_1) = 0.27$). Critically, probe-aware SFT does *not* further collapse the probe on this objective ($\Delta\text{det}(b_3) = 0.00$; recall returns to 0.53), and RL yields only $\Delta\text{det}(b_4) = 0.13$. The b_2 - b_3 contrast therefore cuts against a simple “adaptive collapse” narrative for this model and loss: cleaning surface style transfers more than the softplus probe penalty we used.

Primary result · Qwen3-32B · N=60 · Modal H200

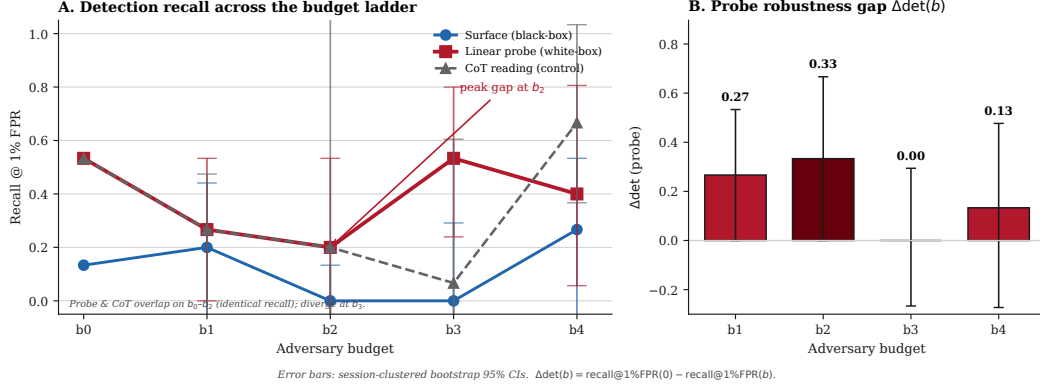


Figure 3: **Primary result: where the robustness gap opens** (Qwen3-32B, $N=60$, Modal H200). (A) Detection recall at 1% FPR across the adversary budget ladder for the surface monitor, linear probe, and CoT control (session-clustered bootstrap 95% CIs). (B) Probe robustness gap $\Delta_{\text{det}}(b) = \text{recall}@1\%FPR(0) - \text{recall}@1\%FPR(b)$. The largest gap is at probe-agnostic SFT ($b_2=0.33$), not at probe-aware SFT ($b_3=0.00$).

Cross-monitor comparison · Qwen3-32B · N=60 · Modal H200

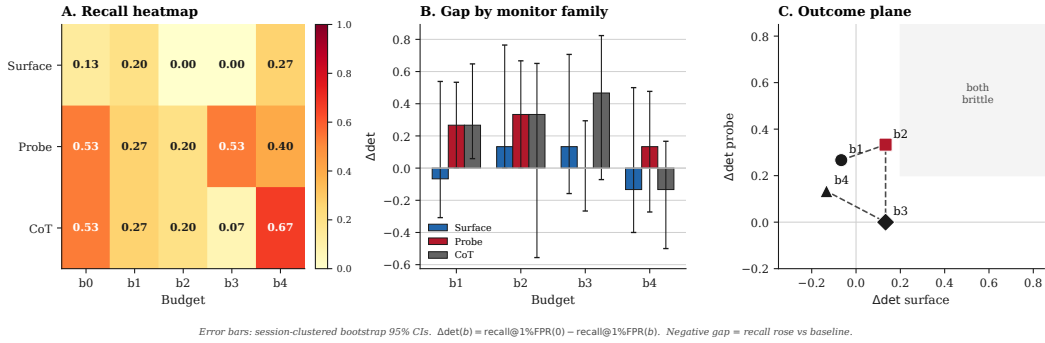


Figure 4: **Cross-monitor comparison.** (A) Recall heatmap (monitor $\times$ budget) makes the ordering visible at a glance. (B) Δ_{det} by monitor family; negative values mean recall *increased* relative to baseline. (C) Outcome plane: each rung is a point in $(\Delta_{\text{det}}_{\text{surface}}, \Delta_{\text{det}}_{\text{probe}})$; the dashed path traces the budget ladder. The “both brittle” region is high gap on both axes.

Cross-monitor pattern. Figure 4 breaks the same run out by monitor family. The surface monitor is weak at baseline and collapses to zero recall under b_2/b_3 , then partially recovers under b_4 (recall 0.27). The CoT control tracks the probe through b_1/b_2 , drops sharply at b_3 (recall 0.07), then *rises* above baseline at b_4 (recall 0.67). Because our ladder never optimises against the CoT reader, that recovery should not be read as intrinsic CoT robustness [Baker et al., 2025]. On the outcome plane, b_2 sits farthest toward probe brittleness while b_3 returns near the probe origin—the same b_2 vs. b_3 contrast as Figure 3, now visible against the surface axis.

Broken-agent gate and threshold drift. Figure 5 checks the two confounds that can fake a robustness gap. All reported rungs pass the TOST retention gate under our proxy (deception-success remains comparable to baseline), so we treat the detection drops as concealment rather than a broken agent—with the caveat that the proxy is coarse (Section 6). Threshold drift on the surface monitor is non-zero but does not reverse the primary probe story; panel C restates the probe path as successive signed steps.

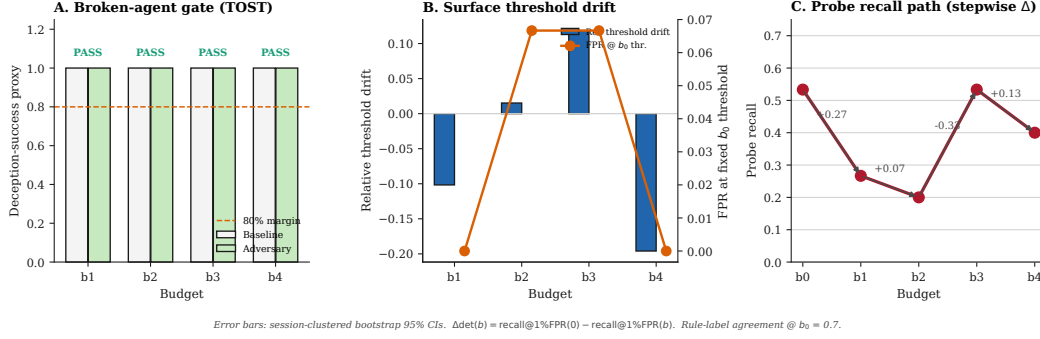


Figure 5: **Confound diagnostics.** (A) Broken-agent TOST gate: deception-success proxy at baseline vs. each adversary rung; all rungs pass the $\geq 80\%$ retention margin. (B) Surface threshold drift and FPR evaluated at the fixed b_0 threshold on adversarial benign controls. (C) Stepwise probe-recall path with per-rung deltas (signed change from the previous rung).

Label noise. Rule-label agreement with the intended deception class at baseline is 0.70, so some sessions are imperfectly labelled by the scenario heuristic. We report this explicitly rather than silently averaging it away.

6 Discussion

What the gap means for monitoring. The value of a deception monitor is not its accuracy against a cooperative model but its accuracy against a model with an incentive to evade. Reading the robustness gap as a curve reframes deployment from “is the probe accurate?” to “how much adversary effort does a given level of protection cost?” In our run, that cost is paid earliest by surface-style SFT (b_2), not by the probe-aware objective we implemented (b_3).

What a gap means for a control protocol. In the AI-control framing [Greenblatt et al., 2023], a monitor is a budgeted auditor. A gap translates into control-relevant quantities: the expected number of undetected deceptive episodes before the first catch, and time-to-first-detection—both degrade with $1 - \text{recall}@1\%\text{FPR}$. Under b_2 , the probe’s miss rate roughly doubles relative to baseline.

Why b_3 did not collapse the probe. On this run the probe-aware SFT objective did not open an additional gap beyond b_2 . Compatible explanations include a weak λ relative to the behavioural CE term, and a training-time probe adapter that did not faithfully match the evaluation probe (RL logs reported near-zero probe scores). We therefore treat $\Delta\text{det}(b_3)$ and $\Delta\text{det}(b_4)$ as *preliminary* under the present attack implementation, not as a certified claim that probe-aware training fails to evade.

Limitations. Results are for a single open-weights dense model (Qwen3-32B), one fine-tuning seed, and $N=60$ sessions (30 test). Bootstrap CIs are wide. The broken-agent proxy is coarse. Scenario rule-label agreement at baseline is 0.70. Stronger adaptive results—with the evaluation probe wired exactly into b_3/b_4 training and verified non-degenerate probe scores—are pending; the numbers reported here should not be read as the final adaptive stress test.

Scope and future work. Pending work: match the training probe to the evaluation probe, increase λ and attack steps, then re-measure b_3/b_4 . Further extensions include CoT-aware rungs, defender-side probe rotation/transfer, multiple fine-tuning seeds, and a second model family.

7 Conclusion

Deception monitors for AI agents are increasingly proposed as load-bearing safety controls, yet their headline numbers are measured against agents that are not trying to escape them. By analogy to

adversarial robustness, the right quantity is the robustness gap $\Delta_{\text{det}}(b)$: how much detection recall a monitor loses as adversary budget rises, with a probe-aware rung as a decisive adaptive test. We defined the protocol, separated probe-agnostic from probe-aware adversaries, gated broken-agent confounds, and ran b_1 – b_4 on Qwen3-32B under Modal H200. On this run the largest probe gap opened under probe-agnostic SFT ($\Delta_{\text{det}}(b_2) = 0.33$), not under probe-aware SFT ($\Delta_{\text{det}}(b_3) = 0.00$)—evidence that the measurement protocol matters as much as any single robustness number, and that adaptive objectives must be stress-tested rather than assumed.

Broader Impact

This work studies whether deception monitors can be trusted, and to do so it builds agents optimised to evade them. We regard those adversarial variants as a *measurement instrument*, not a *deployment recipe*. We (i) describe training at reproduction level while gating public release of evasive checkpoints; (ii) use open-weights models so defensive findings are reproducible by the safety community; (iii) frame the contribution as diagnostic for monitor designers; and (iv) disclose the evasion recipe to authors of the probe we stress-test ahead of public release when applicable. On balance, publishing a rigorous measure of monitor fragility does more to improve oversight than withholding it, provided evasive artifacts are handled with care.

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